AXEL HILS

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https://github.com/shelovesmoussey/generic-work

PROFILE

US and French Citizen with a background in Mathematical Physics. Specialized in applying rigorous physical-mathematical methods and stochastic modeling to complex systems. Ranked 4th in an international Master's program with a focus on high-level theoretical physics and computational modeling.

EDUCATION

Université de Strasbourg Strasbourg, France
Master's Degree (M2) in Pure Mathematics (MF) 2026 – 2027
- **Ranking: TBA**
- **Relevant Coursework:** Stochastic calculus, statistical mechanics, optimal transport theory and probability, stochastic models applied to blockchain-type systems and population dynamics.

Université de Bourgogne Europe Dijon, France
Master's Degree (M2) in Mathematical Physics (Math4Phys) 2025 – 2026
- **Ranking: 4 / 25**
- **Relevant Coursework:** Riemann surfaces, algebraic geometry, Lie groups and algebras, quantum field theory, general relativity, path approach in QFT, Hilbert schemes and moduli spaces.

Master's Degree (M1) in Mathematical Physics (Math4Phys) 2024 – 2025
- **Ranking: 4 / 15**
- **Relevant Coursework:** Stochastic methods, Partial Differential Equations (PDEs), differential geometry, functional analysis, classical mechanics, quantum mechanics.

Bachelor's Degree (L3) in Pure Mathematics 2023 – 2024
- **Ranking: 6 / 50**
- **Relevant Coursework:** Differential calculus, topology, algebra, functional analysis, probability.

Lycée Carnot Dijon Dijon, France
Preparatory Class for Grandes Écoles (CPGE) - PCSI/PC 2021 – 2023
- Two-year intensive program in Mathematics, Physics, and Chemistry preparing for national competitive exams.

EXPERIENCE

Mathematics Teaching (Private Lessons and Tutoring) 2023 – Present
- Mathematics lessons at preparatory-class (CPGE) level and beyond (university level) for over three years, following earlier peer tutoring.
- Supporting students in understanding concepts, exercise methodology, and competitive exam preparation.

SELECTED RESEARCH

Graduate Research Thesis: Lévy Processes in Financial Modeling 2025 – 2026
- Developed mathematical frameworks using Lévy processes to model non-Gaussian market discontinuities and "jump" risks.
- Investigated infinitesimal generators and stochastic differential equations for asset price modeling.

Research Project: Binary Black Hole Solutions via the Ernst Equation 2024 – 2025
- Modeled gravitational dynamics using the Ernst equation within General Relativity.
- Applied numerical methods and complex analysis to identify stable physical solutions for binary systems.

SKILLS

- **Programming:** Python (NumPy, SciPy), C++, SQL, LaTeX, MATLAB (5+ Years).
- **Mathematics:** Stochastic Methods, Functional Analysis, Differential Geometry, Lie Algebras.
- **Physics:** Quantum Field Theory, General Relativity, Classical & Quantum Mechanics.
- **Languages:** English (Native/Bilingual), French (Native/Bilingual), Spanish.

LEISURE & DISTINCTIONS

- **Chess:** Elo 1800 – Advanced pattern recognition and strategic optimization.
- **Interests:** Clash Royale : #235 French player, Weightlifting, Cinema, and thorough reading of scientific literature.